Ex.no:6c	DATA VISUALIZATION USING POWER BI (CITY DAY AIR QUALITY)

Aim:

To clean, preprocess, and transform the City Day air-quality dataset, load the clean data model into Power BI, and build an interactive visual dashboard to analyze AQI levels, pollutant concentrations, city-wise air quality, and trends.

Algorithm:

1. **Import Dataset:** Load city_day.csv into the environment.

2. **Deduplication**: Identify and remove duplicate rows. The uploaded dataset contains 0 duplicate rows.

3. **Handle Missing Values**: Fill missing numeric pollutant/AQI values using the median and categorical values using the mode.

4. **Handle Data Types:** Convert Date to Date format and pollutant/AQI fields to numeric data types.

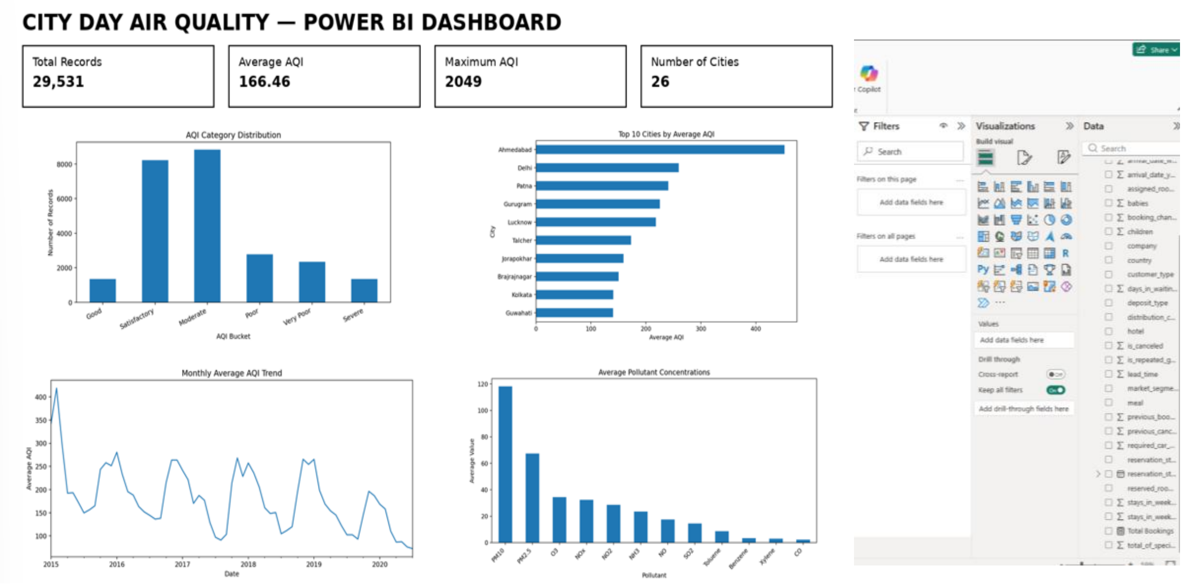
5. **Load Clean Data:** In Power BI Desktop, select Get Data > Text/CSV, choose the cleaned City Day data, and use Transform Data / Apply & Close.

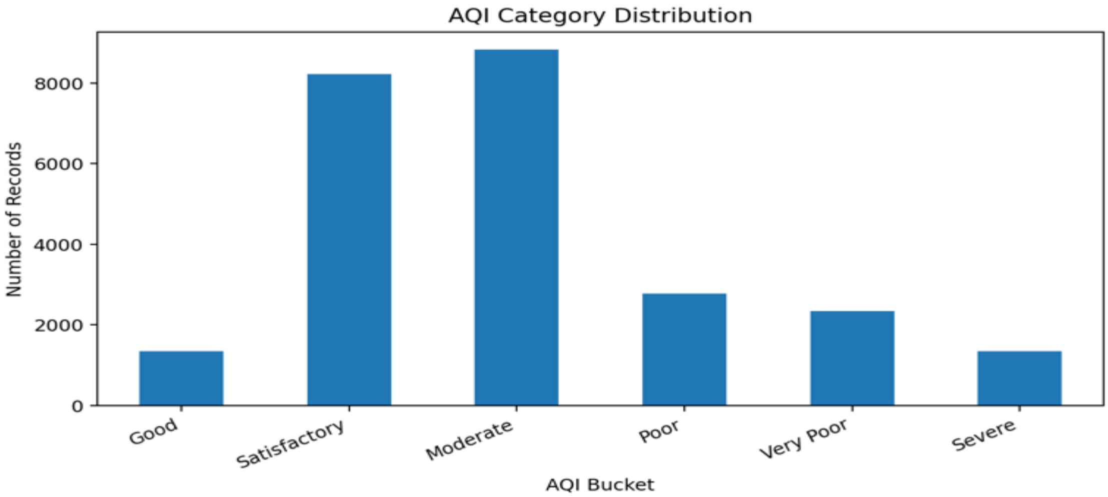
6. **Create Core Measures**: Create measures for Total Records, Average AQI, Maximum AQI, Minimum AQI, and Number of Cities.

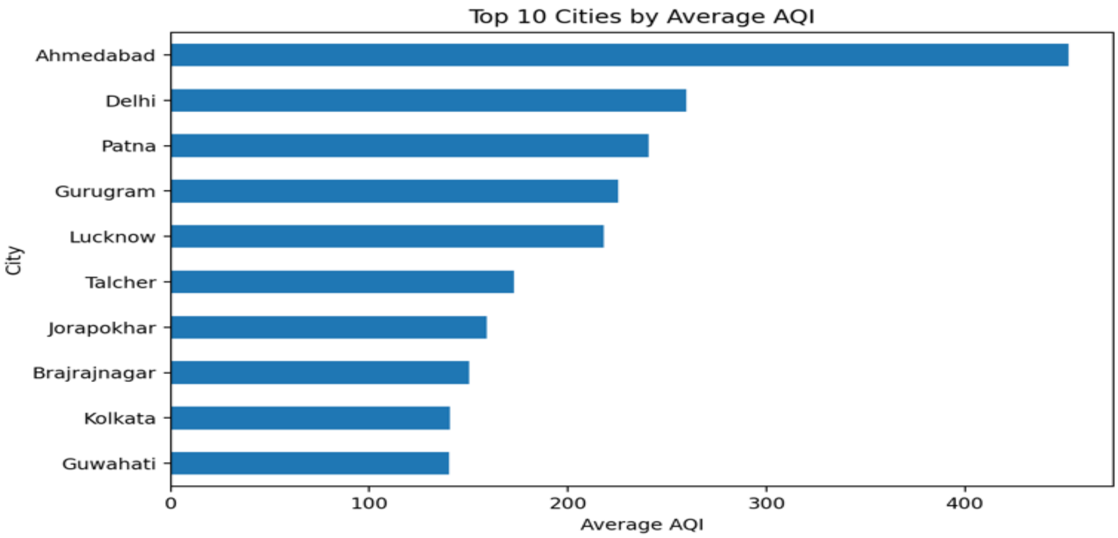
7. **Create Visuals**: Add KPI cards, AQI Bucket distribution, city-wise average AQI, monthly AQI trend, and average pollutant comparison.

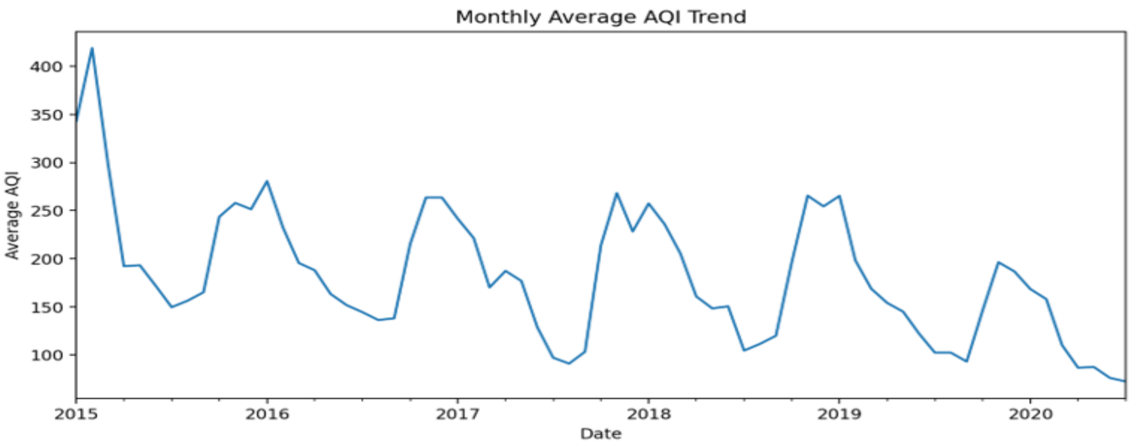
8. **Add Filters**: Add slicers for City, Date, and AQI_Bucket and format the dashboard for interactive analysis.

Output:




AQI Category Distribution
Number of Records
0
2000
4000
6000
8000
Good
Satisfactory
Moderate
Poor
Very Poor
Severe
AQI Bucket


Top 10 Cities by Average AQI
City
Ahmedabad
Delhi
Patna
Gurugram
Lucknow
Talcher
Jorapokhar
Brajrajnagar
Kolkata
Guwahati
0
100
200
300
400
Average AQI


Monthly Average AQI Trend
Average AQI
100
150
200
250
300
350
400
2015
2016
2017
2018
2019
2020
Date

Result: